

The Hong Kong Polytechnic University

Subject Description Form

Subject Code	AAE5208
Subject Title	Satellite Engineering
Credit Value	3
Level	5
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	To equip students with the knowledge, skills, and competencies needed in the space sector, including space environment, orbital mechanics, subsystems and its design, satellite software engineering, integration and test, launch vehicles, ground stations, systems engineering and program management, etc.
Intended Learning Outcomes	Upon completion of this subject, students will be able to: - Demonstrate a thorough understanding of the theories and principles of satellite engineering. - Apply theoretical concepts to analyse and solve practical satellite engineering problems. - Design satellite subsystems and assess its performance via computation and simulation. - Recognise the importance of and exhibit the ability to engage in lifelong learning endeavours.
Subject Synopsis/ Indicative Syllabus	Introduction to Satellite Engineering: Overview of satellite engineering, including its history and the state-of-the-art development, the satellite missions and applications, the mathematical tools and skills used in the satellite engineering course. Space Environment: Space vacuum conditions and the earth atmosphere, the impact of space environment on satellite orbits, attitude, structure and materials, electrostatic charge and discharge, etc. Orbital Mechanics: Newton's laws of motion and gravitation, Kepler's laws, orbit types, orbital transfers and manoeuvres, orbital perturbations. Attitude Determination and Control (ADCS): Coordinate systems, satellite attitude dynamics, actuators and sensors, attitude control systems.

	Telemetry, Tracking and Control Subsystems (TT&C): TT&C functions, command systems, telemetry system, coverage and link analysis, modulation/demodulation. Other Satellite Subsystems: On-board data handling system (OBDH), electric power system, the propulsion system, ground system, etc. Satellite Software Engineering: Software lifecycle, cost, design, test, maintenance; Component-based software engineering Satellite Integration and Test: Facilities and procedures for integration and test, including vibration test, thermal vacuum test, antenna deployment test, electromagnetic compatibility and interference test, etc. Satellite Mission Analysis and Program Management: Mission objectives; Top-level requirements and the flowdown; Preliminary design review (PDR); Interface Control Documents (ICDs), Critical Design Review (CDR); System and mission simulations; Scheduling; Cost; Documentation, etc. Nanosatellite design project: Nanosatellite mission analysis and design with the NASA GMAT software and/or the CubeSat Toolbox for MATLAB®																								
Teaching/Learning Methodology	The teaching and learning methods include lectures, project and homework assignments. Technical/practical examples and problems will be raised and discussed in class. The nanosatellite project is designed to enhance students’ learning and understanding of satellite missions. <table><tr><th rowspan="2">Teaching/Learning Methodology</th><th colspan="4">Outcomes</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th></tr><tr><td>Lecture</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>Project</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>Homework assignments</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr></table>	Teaching/Learning Methodology	Outcomes				a	b	c	d	Lecture	✓	✓	✓	✓	Project	✓	✓	✓	✓	Homework assignments	✓	✓	✓	✓
Teaching/Learning Methodology	Outcomes																								
	a	b	c	d																					
Lecture	✓	✓	✓	✓																					
Project	✓	✓	✓	✓																					
Homework assignments	✓	✓	✓	✓																					

Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks		% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)			
				a	b	c	d
	1. Homework assignments		25%	✓	✓	✓	✓
	2. Project		25%	✓	✓	✓	✓
	3. Final examination		50%	✓	✓	✓	
	Total		100%				
	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: 0.5 × Continuous Assessment + 0.5 × Final Examination Continuous assessment, comprising homework assignments and project report, functions to track students' advancement, facilitate self-evaluation against intended learning goals, and foster the amalgamation of knowledge gained from lectures. Conversely, the final examination objectively assesses students' knowledge acquisition, capacity to understand and critically analyse problems independently, and overall attainment of subject learning objectives.						
Student Study Effort Expected	Class contact:						
	▪ Lecture					30 Hrs.	
	▪ Lab					9 Hrs.	
	Other student study effort:						
	▪ Self-study					78 Hrs.	
	Total student study effort					117 Hrs.	
Reading List and References	1. Spacecraft Systems Engineering, 4th ed., 2011, P Fortescue <i>et al.</i> , Wiley 2. Low Earth Orbit Satellite Design, 1st ed., 2018, EG Sebestyen <i>et al.</i> , Springer 3. Space Vehicle Dynamics and Control, 2nd ed., 2008, B Wie, AIAA						